

CHENNA KESAVA REDDY YENUGU

Embedded Systems Engineer - Trainee

Bangalore, India | +91-8074453995

chennakesava0517@gmail.com ♦ linkedin.com/in/chenna-kesava-b47901228

SUMMARY

Embedded Systems Engineer with hands-on experience in Embedded C programming, ARM microcontrollers through advanced training and projects. Skilled in Firmware Development, peripheral driver implementation, and communication protocols including CAN, SPI, UART, and I2C. Strong foundation in automotive ECU Communication, Embedded debugging, and real-time embedded systems development.

EDUCATION

Bachelor of Technology (B.Tech) - Electrical and Electronics Engineering (Percentage: 78.2%)

JNTUA College of Engineering, Kalikiri

2021-2025

SKILLS

Programming Languages: C, Embedded C, C++

Embedded Systems: ARM7 Microcontroller, LPC2129, Peripheral Driver Development, Firmware Development

Networking & Communication: CAN, UART, SPI, I2C, GPIO, TCP/IP, UDP, Raw Sockets, Socket Programming

Operating Systems: Linux Internals, Linux System Programming, FreeRTOS (Basic Knowledge)

Debugging Tools: GDB, Embedded System Debugging

EXPERIENCE

Embedded Systems Trainee

06/2025 – 01/2026

Vector India Pvt. Ltd., Bangalore

Automotive ECU Communication using CAN protocol - Hands-on Training

- Developed and validated automotive ECU-to-ECU communication logic for vehicle control functions
- Implemented CAN message flow between ECUs for automotive indicators such as headlight and turn signals
- Designed control logic and validated system behavior under multiple test scenarios
- Executed Debugging, integration testing, and functional verification in real-time environment

PROJECTS

Smart Vehicle Parameter Monitoring system using High-Speed SPI and I2C Communication

- Implemented a real-time vehicle parameter monitoring system by interfacing multiple sensors using high-speed SPI and I2C protocols on an embedded microcontroller
- Developed efficient data acquisition and processing algorithms to monitor vehicle metrics with impact accuracy

Intelligent Ultrasonic-Based Reverse Parking and Collision Avoidance System using CAN protocol

- Designed a real-time automotive safety system using ultrasonic sensors with Ipc2129 MCU for accurate obstacle distance
- Developed a collision prevention and reverse parking guidance system using Ultrasonic sensors
- Implemented Embedded C algorithms for proximity detection and adaptive alerts, enabling reliable reverse parking

Automated Employee Attendance System using RFID technology

- Designed and developed a RFID-based real-time attendance system using LPC2129 ARM7 microcontroller to automate employee entry and exit tracking
- Used UART & SPI communication protocol to interface RFID reader, ensured reliable tag detection and data acquisition
- Developed Embedded C Firmware for Unique ID verification, attendance logging and system control with optimized memory usage

Home Automation using Bluetooth (Arduino)

- Developed a Bluetooth based home automation system using Arduino to remotely control household appliances through a mobile application
- Implemented wireless communication and relay interfacing for real-time device switching, Improving convenience.